

Class 19: Pertussis Mini Project

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Background

Pertissis is a bacterial lung infection also known as whooping cough. Let’s begin by examining CDC reported case numbers in the US:

```
cdc <- data.frame(  
  year = c(1922L,1923L,1924L,1925L,  
           1926L,1927L,1928L,1929L,1930L,1931L,  
           1932L,1933L,1934L,1935L,1936L,  
           1937L,1938L,1939L,1940L,1941L,1942L,  
           1943L,1944L,1945L,1946L,1947L,  
           1948L,1949L,1950L,1951L,1952L,  
           1953L,1954L,1955L,1956L,1957L,1958L,  
           1959L,1960L,1961L,1962L,1963L,  
           1964L,1965L,1966L,1967L,1968L,1969L,  
           1970L,1971L,1972L,1973L,1974L,  
           1975L,1976L,1977L,1978L,1979L,1980L,  
           1981L,1982L,1983L,1984L,1985L,  
           1986L,1987L,1988L,1989L,1990L,  
           1991L,1992L,1993L,1994L,1995L,1996L,  
           1997L,1998L,1999L,2000L,2001L,  
           2002L,2003L,2004L,2005L,2006L,2007L,  
           2008L,2009L,2010L,2011L,2012L,  
           2013L,2014L,2015L,2016L,2017L,2018L,
```

```

2019L,2020L,2021L,2022L,2023L,2024L),
cases = c(107473,164191,165418,152003,
202210,181411,161799,197371,
166914,172559,215343,179135,265269,
180518,147237,214652,227319,103188,
183866,222202,191383,191890,109873,
133792,109860,156517,74715,69479,
120718,68687,45030,37129,60886,
62786,31732,28295,32148,40005,
14809,11468,17749,17135,13005,6799,
7717,9718,4810,3285,4249,3036,
3287,1759,2402,1738,1010,2177,2063,
1623,1730,1248,1895,2463,2276,
3589,4195,2823,3450,4157,4570,
2719,4083,6586,4617,5137,7796,6564,
7405,7298,7867,7580,9771,11647,
25827,25616,15632,10454,13278,
16858,27550,18719,48277,28639,32971,
20762,17972,18975,15609,18617,
6124,2116,3044,7063,22538)
)

```

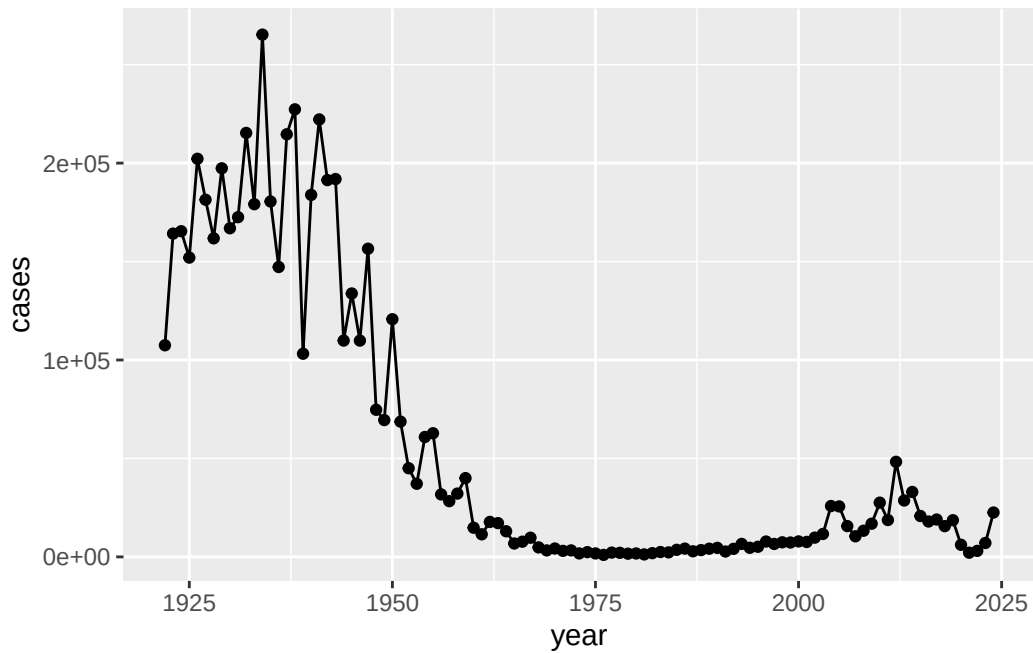
Plot of cases per year for Pertussis in the US

```

library(ggplot2)

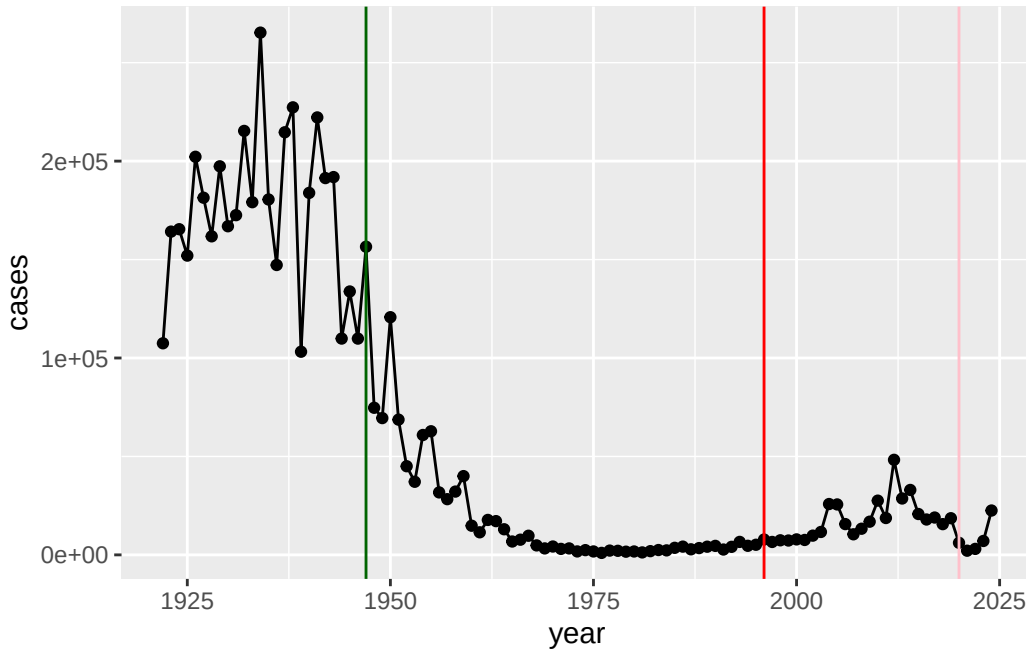
ggplot(cdc)+
  aes(year, cases)+
  geom_point()+
  geom_line()

```



Add some major milestone timepoints to our plot:

```
ggplot(cdc)+  
  aes(year, cases)+  
  geom_point()+  
  geom_line()+  
  geom_vline(xintercept = 1947, col="darkgreen")+  
  geom_vline(xintercept = 1996, col="red")+  
  geom_vline(xintercept = 2020, col="pink")
```



The full introduction of mandatory wP (whole-cell) Pertussis immunization in the mid 1940s lead to a dramatic reduction in case numbers (from over 200,000 to 100s)

The switch to the aP (newer acellular) formalization in 1996 coincides with a gradual increase in case numbers over the next two decades.

The 2020 lockdowns and social distancing measures implemented to combat the COVID-19 pandemic lead to a dramatic drop in Pertussis cases, but case numbers have rebounded in the last few years.

The CMI-PB project

The mission of CMI-PB is to provide the scientific community with a comprehensive, high-quality and freely accessible resource of Pertussis booster vaccination.

Website: <https://www.cmi-pb.org/>

The make their data available via JSON format API endpoints - basically the database tables in a key:value type format like "infancy_vac": "wP". To read this we can use the `read_json` function from the `jsonlite` package. Install with `install.packages("jsonlite")`

```
library(jsonlite)

subject <- read_json(path="http://cmi-pb.org/api/v5_1/subject", simplifyVector = TRUE)
```

```
head(subject)
```

	subject_id	infancy_vac	biological_sex	ethnicity	race
1	1	wP	Female	Not Hispanic or Latino	White
2	2	wP	Female	Not Hispanic or Latino	White
3	3	wP	Female	Unknown	White
4	4	wP	Male	Not Hispanic or Latino	Asian
5	5	wP	Male	Not Hispanic or Latino	Asian
6	6	wP	Female	Not Hispanic or Latino	White

	year_of_birth	date_of_boost	dataset
1	1986-01-01	2016-09-12	2020_dataset
2	1968-01-01	2019-01-28	2020_dataset
3	1983-01-01	2016-10-10	2020_dataset
4	1988-01-01	2016-08-29	2020_dataset
5	1991-01-01	2016-08-29	2020_dataset
6	1988-01-01	2016-10-10	2020_dataset

Q. how many “subjects”/individuals are in this dataset?

```
nrow(subject)
```

```
[1] 172
```

Q. How many wP and aP subjects are there?

```
table(subject$infancy_vac)
```

```
aP wP  
87 85
```

Q. What is the breakdown by “biological_sex” and “race”?

```
table(subject$biological_sex)
```

```
Female  Male  
112    60
```

```
table(subject$race)
```

```

American Indian/Alaska Native
      1
Asian
     44
Black or African American
      5
More Than One Race
     19
Native Hawaiian or Other Pacific Islander
      2
Unknown or Not Reported
     21
White
     80

```

```
table(subject$race, subject$biological_sex)
```

	Female	Male
American Indian/Alaska Native	0	1
Asian	32	12
Black or African American	2	3
More Than One Race	15	4
Native Hawaiian or Other Pacific Islander	1	1
Unknown or Not Reported	14	7
White	48	32

This breakdown is not particularly representative of the US population - this is a serious caveat for this study. However, it is still the largest sample of its type ever assembled.

```
specimen <- read_json(path="http://cmi-pb.org/api/v5_1/specimen", simplifyVector = TRUE)
```

```
ab_titer <- read_json(path="http://cmi-pb.org/api/v5_1/plasma_ab_titer", simplifyVector = TRUE)
```

```
head(specimen)
```

	specimen_id	subject_id	actual_day_relative_to_boost	
1	1	1	-3	
2	2	1	1	
3	3	1	3	
4	4	1	7	
5	5	1	11	
6	6	1	32	

	planned_day_relative_to_boost	specimen_type	visit
1	0	Blood	1
2	1	Blood	2
3	3	Blood	3
4	7	Blood	4
5	14	Blood	5
6	30	Blood	6

We need to “join” or link these tables with the `subject` table so we can begin to analyze this data and know who a give Ab sample was collected for and when

```
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

`filter`, `lag`

The following objects are masked from 'package:base':

`intersect`, `setdiff`, `setequal`, `union`

```
meta <- inner_join(subject, specimen)
```

Joining with `by = join\_by(subject\_id)`

```
head(meta)
```

	subject_id	infancy_vac	biological_sex	ethnicity	race
1	1	wP	Female Not Hispanic or Latino	White	
2	1	wP	Female Not Hispanic or Latino	White	
3	1	wP	Female Not Hispanic or Latino	White	
4	1	wP	Female Not Hispanic or Latino	White	
5	1	wP	Female Not Hispanic or Latino	White	
6	1	wP	Female Not Hispanic or Latino	White	

	year_of_birth	date_of_boost	dataset	specimen_id
1	1986-01-01	2016-09-12	2020_dataset	1
2	1986-01-01	2016-09-12	2020_dataset	2
3	1986-01-01	2016-09-12	2020_dataset	3
4	1986-01-01	2016-09-12	2020_dataset	4
5	1986-01-01	2016-09-12	2020_dataset	5
6	1986-01-01	2016-09-12	2020_dataset	6

	actual_day_relative_to_boost	planned_day_relative_to_boost	specimen_type
1	-3	0	Blood
2	1	1	Blood
3	3	3	Blood
4	7	7	Blood
5	11	14	Blood
6	32	30	Blood

	visit
1	1
2	2
3	3
4	4
5	5
6	6

Now let's join the `ab_titer` with our `meta` table so we have all the information in one place about a given Ab measurement

```
ab_data <- inner_join(meta, ab_titer)
```

Joining with ``by = join_by(specimen_id)``

```
head(ab_data)
```

	subject_id	infancy_vac	biological_sex	ethnicity	race
1	1	wP	Female Not Hispanic or Latino	White	
2	1	wP	Female Not Hispanic or Latino	White	

3	1	wP	Female Not Hispanic or Latino White				
4	1	wP	Female Not Hispanic or Latino White				
5	1	wP	Female Not Hispanic or Latino White				
6	1	wP	Female Not Hispanic or Latino White				
	year_of_birth	date_of_boost	dataset	specimen_id			
1	1986-01-01	2016-09-12	2020_dataset	1			
2	1986-01-01	2016-09-12	2020_dataset	1			
3	1986-01-01	2016-09-12	2020_dataset	1			
4	1986-01-01	2016-09-12	2020_dataset	1			
5	1986-01-01	2016-09-12	2020_dataset	1			
6	1986-01-01	2016-09-12	2020_dataset	1			
	actual_day_relative_to_boost	planned_day_relative_to_boost	specimen_type				
1		-3	0	Blood			
2		-3	0	Blood			
3		-3	0	Blood			
4		-3	0	Blood			
5		-3	0	Blood			
6		-3	0	Blood			
	visit	isotype	is_antigen_specific	antigen	MFI	MFI_normalised	unit
1	1	IgE	FALSE	Total	1110.21154	2.493425	UG/ML
2	1	IgE	FALSE	Total	2708.91616	2.493425	IU/ML
3	1	IgG	TRUE	PT	68.56614	3.736992	IU/ML
4	1	IgG	TRUE	PRN	332.12718	2.602350	IU/ML
5	1	IgG	TRUE	FHA	1887.12263	34.050956	IU/ML
6	1	IgE	TRUE	ACT	0.10000	1.000000	IU/ML
	lower_limit_of_detection						
1	2.096133						
2	29.170000						
3	0.530000						
4	6.205949						
5	4.679535						
6	2.816431						

How many Ab measurements do we have in total?

```
nrow(ab_data)
```

```
[1] 61956
```

Q. How many different isotypes (type of Ab) are in the dataset?

```
unique(ab_data$isotype)
```

```
[1] "IgE" "IgG" "IgG1" "IgG2" "IgG3" "IgG4"
```

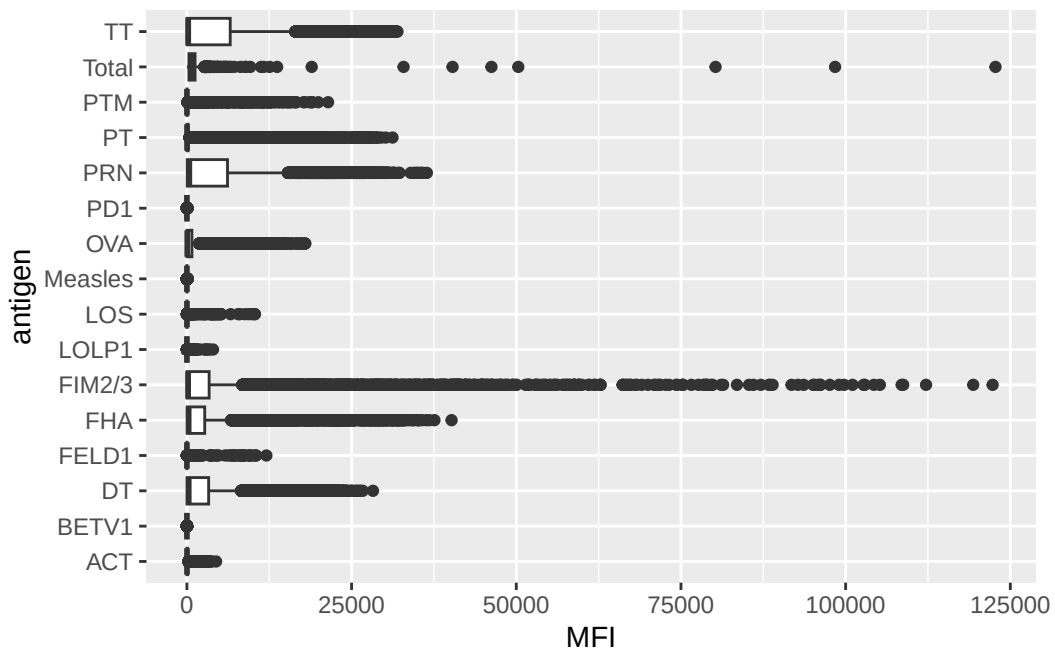
Q. How many different antigens?

```
unique(ab_data$antigen)
```

```
[1] "Total" "PT" "PRN" "FHA" "ACT" "LOS" "FELD1"  
[8] "BETV1" "LOLP1" "Measles" "PTM" "FIM2/3" "TT" "DT"  
[15] "OVA" "PD1"
```

```
ggplot(ab_data)+  
  aes(MFI, antigen)+  
  geom_boxplot()
```

Warning: Removed 1 row containing non-finite outside the scale range (`stat\_boxplot()`).

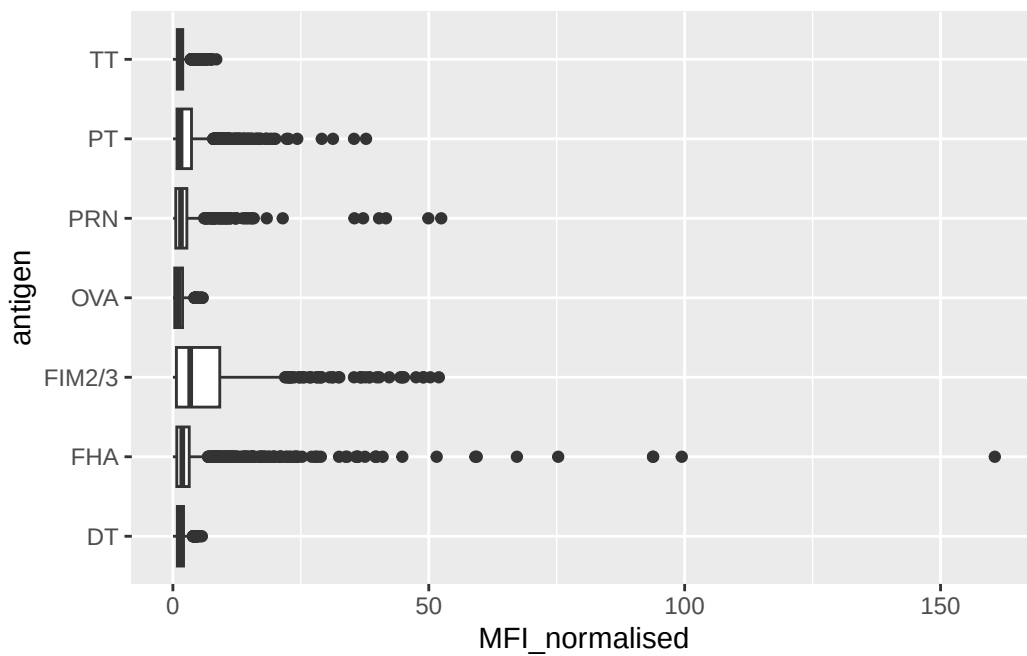


Examine IgG Ab titer levels

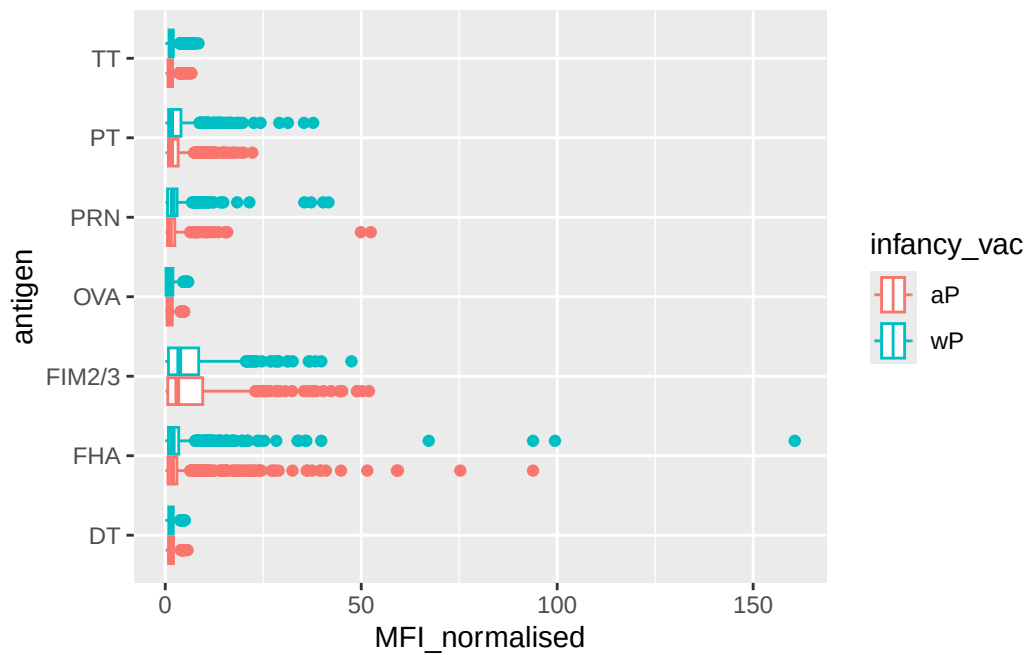
IgG is crucial for long-term immunity and responding to bacterial & viral infections

```
igg <- ab_data |>
  filter(isotype=="IgG")
```

```
ggplot(igg)+
  aes(MFI_normalised, antigen)+
  geom_boxplot()
```

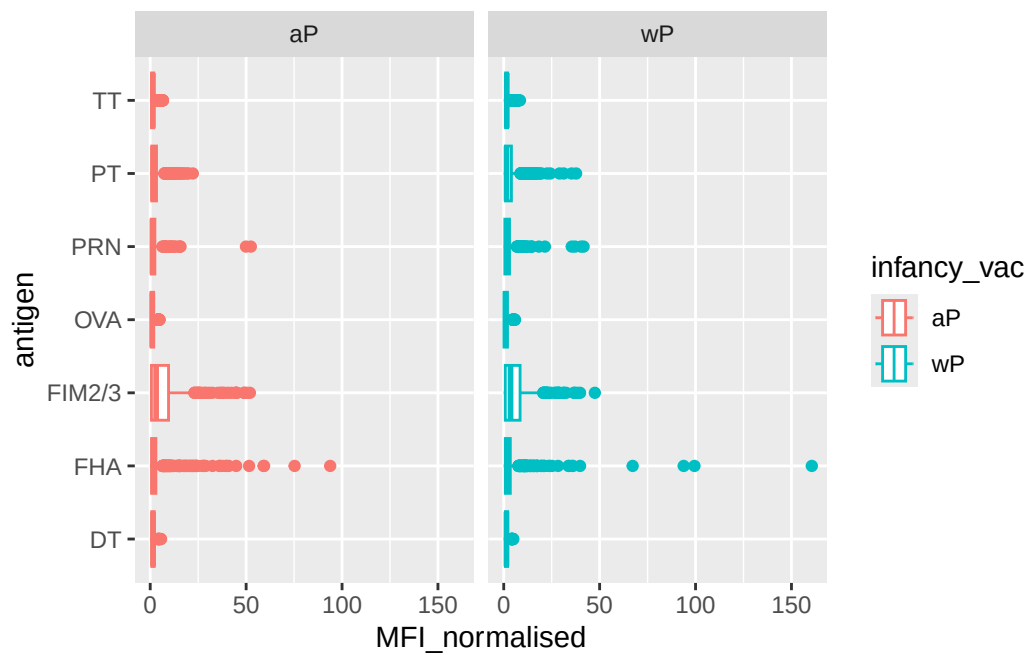


```
ggplot(igg)+
  aes(MFI_normalised, antigen, col=infancy_vac)+
  geom_boxplot()
```



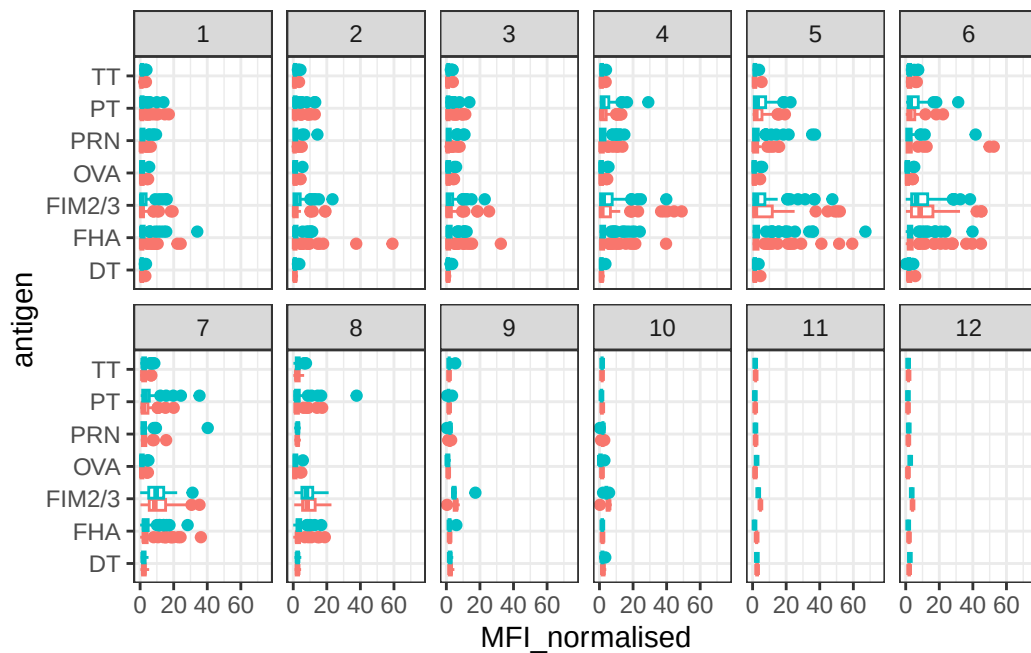
We can “facet” our plot by wP vs aP

```
ggplot(igg)+
  aes(MFI_normalised, antigen, col=infancy_vac)+
  geom_boxplot()+
  facet_wrap(~infancy_vac)
```



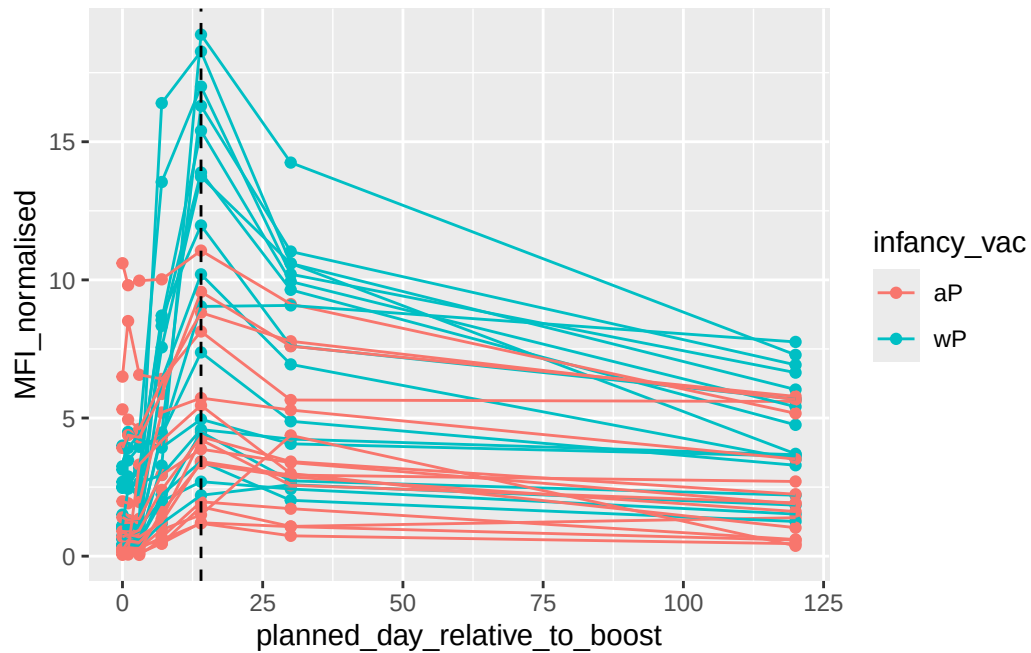
```
ggplot(igg) +
  aes(MFI_normalised, antigen, col=infancy_vac ) +
  geom_boxplot(show.legend = FALSE) +
  facet_wrap(vars(visit), nrow=2) +
  xlim(0,75) +
  theme_bw()
```

Warning: Removed 5 rows containing non-finite outside the scale range (`stat\_boxplot()`).



More advance analysis digging into individual antigen responses over time:

```
filter(igg, antigen=="PT", dataset=="2021_dataset") %>%
  ggplot() +
  aes(x=planned_day_relative_to_boost,
       y=MFI_normalised,
       col=infancy_vac,
       group=subject_id) +
  geom_point() +
  geom_line() +
  geom_vline(xintercept=14, linetype="dashed")
```



This plot shows the time course of Pertussis toxin (PT) antibody responses for large set of wP (teal color) and aP (read color) individuals. Levels peak at day 14 and are larger in magnitude for wP than aP in individuals.

There are lots of cool things to explore in this dataset and we need coding and biology knowledge to do it effectively